Course Name:	Web Development Laboratory	**Semester:**	**III**
Date of Performance:	___ / ___ / ______	**DIV/ Batch No:**	
Student Name:		**Roll No:**	

Experiment No: 1

Title

Design and Development of an AI-based Smart Irrigation Advisory Web Portal Front-End using HTML5

Aim of the Experiment

To design and develop a multi-page, use-case-based web portal front-end using HTML5 for an AI-based Smart Irrigation Advisory System for Sugarcane Crop (KJS-AGR-01), demonstrating proper HTML5 document structure, semantic elements, formatting, lists, hyperlinks, images, image maps, tables, forms, and multimedia elements.

Objectives of the Experiment

- To develop structured and semantic HTML5 web pages for an AI-enabled smart irrigation advisory system.
- To create interactive web pages for farmer registration, irrigation recommendations, weather information, crop monitoring, and advisory services.
- To apply a real-world AI and IoT-based agriculture use case (Smart Irrigation Advisory System) as the context for front-end web development.
- To build a complete HTML5 front-end portal that can later be enhanced using CSS3, JavaScript, PHP, and React.

COs to be achieved

CO1: Developing webpages using HTML and CSS.

Books / Journals / Websites references

1. *(Students should write — e.g., MDN Web Docs HTML5 reference, W3C.)*

Theory

HTML5 is the latest version of HyperText Markup Language used to create structured and interactive web pages. It introduces semantic elements such as <header>, <nav>, <main>,

<section>, <article>, and <footer>, making web pages more organized, accessible, and search-engine friendly.

HTML5 also provides support for multimedia elements like <audio> and <video>, interactive forms, tables, hyperlinks, image maps, and embedded content through <iframe>. These features help developers create user-friendly web applications for various domains such as healthcare, education, business, and agriculture.

In this experiment, HTML5 is used to design the front-end of an AI-based Smart Irrigation Advisory System. The portal enables farmers to register their farms, view irrigation schedules, monitor weather conditions, access crop information, receive AI-generated irrigation recommendations, and visualize farm locations. The application serves farmers, agricultural officers, researchers, and sugar factory management by providing organized and easy-to-understand agricultural information.

Problem Statement

"AI-based Smart Irrigation Advisory Web Portal"

An agricultural organization wants to develop a basic HTML5-based web portal front-end for an **AI-based Smart Irrigation Advisory System**. The portal should display farm information, soil moisture status, weather forecasts, irrigation recommendations, crop information, and advisory services for farmers, agricultural officers, researchers, and sugar factory management.

Each task below contributes one page or feature of this portal, representing different modules of the Smart Irrigation Advisory System.

Task 1: Create the Homepage Structure

Scenario

Design the homepage of the **Irrigation Advisory System**.

Requirements

- Use proper HTML5 document structure tags:
  - <header>, <nav>, <section>, <article>, <aside>, <footer>
- Header must contain:
  - System Name in <h1>
  - Tagline: *"Smart Irrigation using AI & Sensor Technology"*
- Navigation menu with links:
  - Home
  - Farm Dashboard
  - Irrigation Advisory
  - Sensor Data

    - Contact
- Footer with copyright information.

Concepts Tested

- HTML5 document structure
- Semantic tags
- Headings
- Hyperlinks

Deliverable: index.html

Task 2: Display Farm Information using Lists

Scenario

Display important farm information and irrigation resources.

Requirements

Create:

- Ordered List for
    - Irrigation Workflow
- Unordered List for
    - Sensors Used
        - Soil Moisture Sensor
        - Temperature Sensor
        - Humidity Sensor
        - Rainfall Sensor
- Description List for
    - Crop Stages
        - Germination
        - Tillering
        - Grand Growth
        - Maturity

Use formatting tags:

- <b>
- <i>
- <mark>

Concepts Tested

- Lists

- Formatting tags
- Text formatting

Deliverable: farm-info.html

Task 3: Irrigation Advisory Page using Formatting Tags

Scenario

Display today's irrigation recommendation for a sugarcane field.

Requirements

Display:

- Plot ID
- Farmer Name
- Soil Moisture
- Recommended Irrigation Duration
- Water Stress Level

Use:

- <strong>
- <em>
- <mark>
- <small>
-
- <hr>

Concepts Tested

- Text formatting
- Block-level elements
- Inline formatting

Deliverable: advisory.html

Task 4: Farm Gallery using Images

Scenario

Create a gallery showing different farm conditions.

Requirements

Display four images:

- Healthy Sugarcane Crop
- Drip Irrigation System
- Soil Moisture Sensor
- Weather Monitoring Station

Each image should:

- Open detailed information when clicked
- Include:
    - alt
    - title
    - width
    - height

Use:

- <figure>
- <figcaption>

Concepts Tested

- Images
- Hyperlinks
- Global attributes

Deliverable: gallery.html

Task 5: Farm Plot Navigation using Image Map

Scenario

Provide navigation to different farm plots.

Requirements

Display a farm map.

Create clickable areas for:

- Plot A
- Plot B
- Plot C

Each area should open the corresponding plot details.

Use:

- <map>

- <area>

Concepts Tested

- Image Maps
- Hyperlinks

Deliverable: plot-map.html

Task 6: Soil Moisture & Irrigation Table

Scenario

Display irrigation recommendations for multiple plots.

Requirements

Create a table with columns:

- Plot ID
- Farmer Name
- Soil Moisture (%)
- Weather
- Irrigation Recommendation
- Water Requirement

Use

- <caption>
- <thead>
- <tbody>
- <tfoot>
- rowspan
- colspan

Apply suitable formatting.

Concepts Tested

- HTML Tables
- Table Attributes

Deliverable: irrigation-table.html

Task 7: Farmer Registration Form

Scenario

Create a registration page for farmers.

Requirements

Use form controls:

- Name
- Mobile Number
- Email
- Plot ID
- Village
- Crop Stage
- Soil Type
- Irrigation Method
- Date
- File Upload (Farm Photo)

Include

- <label>
- <fieldset>
- <legend>

Use HTML5 attributes

- required
- placeholder
- pattern
- autofocus

Concepts Tested

- HTML Forms
- HTML5 Input Types
- Validation Attributes

Deliverable: registration.html

Task 8: Display Weather Forecast using IFrame

Scenario

Display weather information without leaving the website.

Requirements

Embed

- IMD Weather Forecast page
- Google Maps showing the farm location

Use

- <iframe>

Place inside a section called

Today's Weather Forecast

Concepts Tested

- Inline Frames
- External Content Integration

Deliverable: weather.html

Task 9: Add Multimedia & Executable Content

Scenario

Create a multimedia awareness page for farmers.

Requirements

Add

- A video explaining Smart Irrigation
- Background audio explaining water conservation
- JavaScript alert

Welcome to the Irrigation Advisory System

Use

- <video>
- <audio>
- <script>

Enable

- controls
- autoplay
- loop

Concepts Tested

- Multimedia Tags
- JavaScript
- Executable Content

Deliverable: media.html

Code :

Output / Screenshot:

Conclusion and Discussion:
(students should write in your words) (Also take feedback from fellow classmates and faculty to add here)